

Miguel Espinosa

AI RESEARCHER

✉ mespinosami@gmail.com | 🏠 miguel-espinosa.github.io | 🔗 miguel-espinosa | 🎓 Google Scholar | in miguel-espinosa

My research lies in the intersection of Computer Vision and Earth Observation. I am interested in diffusion models, self-supervised methods for data fusion, representation learning, and adapting foundational models for large domain shifts.

Education

PhD in Computer Vision for Earth Observation

Edinburgh, UK

UNIVERSITY OF EDINBURGH

October 2022 - October 2026

- *Foundation Models for Earth Observation* (supervised by Elliot J. Crowley), [SENSE CDT](#).

MSc in Artificial Intelligence

Madrid, Spain

POLYTECHNIC UNIVERSITY OF MADRID · 1ST CLASS HONOURS (~ 93%)

September 2021 - June 2022

- **Research** in *Computer Vision and Aerial Robotics* department: “Facial landmarks detection with deep learning”.

BSc in Computer Science (Bilingual)

Madrid, Spain

UNIVERSITY CARLOS III OF MADRID · 1ST CLASS HONOURS (~ 87%)

September 2017 - May 2021

- **BSc project:** “Self-awareness in a UAV swarm for the complete coverage of its surroundings” (97%)

Experience

European Space Agency, ESA

ESRIN, Frascati, Rome

VISITING RESEARCHER

January 2025 - April 2025

- *Research on multi-modal generative modelling of Copernicus data*
- Development of COP-GEN Beta, a generative diffusion model for zero-shot translation between optical, radar, and elevation data from the Major TOM dataset. Working on multi-modal representation learning and model evaluation.

Canon Medical Research Europe Ltd.

Edinburgh

AI RESEARCH INTERN

June 2022 - August 2022

- *Research in NLP for Clinical Temporal Relation Extraction.*
- Data analysis, exploration, design and implementation of ML models for the extraction of temporal relations from non-structured clinical text.

MeVita

Oxford (remote)

ALGORITHM DEVELOPER INTERN

July 2021 - September 2021

- *Research in ML for natural language (NLP) to solve open-ended problems.*
- Optimisation of the address detection and redaction in the CV pipeline. From research to idea to efficient implementation in C# into production. Received close mentorship from experienced professionals.

Huawei Technologies R&D (UK) Ltd.

Edinburgh Research Center (remote)

RESEARCH INTERN

June 2020 - September 2020

- *Responsible for analysis of the DL framework built on Julia programming language.*
- Contribution to MindSpore DL framework and its integration with Julia as front-end language. Working on source-to-source code generation for the forward and backwards pass with automatic differentiation.

Publications

COP-GEN: Latent Diffusion Transformer for Copernicus Earth Observation Data

[pdf] [code]

ARXIV PREPRINT 2026

Miguel Espinosa*, Eva Gmelich Meijling, Valerio Marsocci, Elliot Crowley, Mikolaj Czerkawski

COP-GEN-Beta: Unified Generative Modelling of COPernicus Imagery Thumbnails

[pdf] [code]

CVPR MORSE WORKSHOP 2025

Miguel Espinosa*, Valerio Marsocci, Yuru Jia, Elliot Crowley, Mikolaj Czerkawski

No time to train! Training-Free Reference-Based Instance Segmentation

[\[pdf\]](#) [\[code\]](#)

ARXIV PREPRINT 2025

Miguel Espinosa*, Chenhongyi Yang*, Linus Ericsson, Steven McDonagh, Elliot J Crowley

There is no samantics! exploring sam as a backbone for visual understanding tasks

[\[pdf\]](#) [\[code\]](#)

ARXIV PREPRINT 2024

Miguel Espinosa*, Chenhongyi Yang*, Linus Ericsson, Steven McDonagh, Elliot J Crowley

einspace: Searching for neural architectures from fundamental operations

[\[pdf\]](#) [\[code\]](#)

NEURIPS 2024

Linus Ericsson*, **Miguel Espinosa**, Chenhongyi Yang, Antreas Antoniou, Amos J Storkey, Shay Cohen, Steven McDonagh, Elliot J Crowley

PlainMamba: Improving non-hierarchical mamba in visual recognition

[\[pdf\]](#) [\[code\]](#)

BMVC 2024

Chenhongyi Yang*, Zehui Chen*, **Miguel Espinosa***, Linus Ericsson, Zhenyu Wang, Jiaming Liu, Elliot J Crowley

Generate Your Own Scotland: Satellite Image Generation Conditioned on Maps

[\[pdf\]](#) [\[code\]](#)

NEURIPS 2023 WORKSHOP ON DIFFUSION MODELS

Miguel Espinosa*, Elliot J Crowley

Academic

Teaching

UNIVERSITY OF EDINBURGH

- *Data Analysis and Machine Learning* (2024)
- *Programming Skills for Engineers* (2025)

Reviewing

- **Conferences:** ICML Silver Award (2026), ICLR (2026), ICCV (2025), CVPR (2025), ICLR (2025), BMVC Outstanding Reviewer (2024), ECCV (2024), BMVC (2023)
- **Workshops:** ICML TerraBytes (2025), CVPR MORSE (2025), NeurIPS Diffusion Models (2023), BMVC ML for EO (2023)

Presenting

- **Conferences:** EGU (2026), BMVC (2025), NeurIPS Workshop (2024)
- **Events:** ESA Science Hub Challenge September 2025 (Frascati 2025), ESA-NASA International Workshop on AI Foundation Model for Earth Observation (Frascati 2025), ESA Earth Observation and Remote Sensing Workshop (ESEC-Galaxia Belgium 2024), ELLIS Summer School on Probabilistic ML (Cambridge 2023), Data Study Group Alan Turing Institute (London 2023)
- **Talks:** EGU Oral Presentation (COP-GEN: Stochastic Generative Modelling of Copernicus Data)
IQ Capital tech talk (Advancing Large Vision and Generative Models for Visual Understanding & EO)

Awards

Jose Cuenca Excellence Award

Madrid, Spain

POLYTECHNIC UNIVERSITY OF MADRID

April 2022

Research Fellowship

Madrid, Spain

POLYTECHNIC UNIVERSITY OF MADRID

January 2022 - May 2022

Excellence Grant (x2)

Madrid, Spain

COMUNIDAD DE MADRID

2019-2020 | 2020-2021

Languages

Spanish, Catalan, English

Technical Skills

- | | |
|-------------------|--|
| Languages | Python (Numpy, Matplotlib, Scikit-learn, Pandas), Java, C++, Shell |
| Frameworks | PyTorch, Tensorflow |
| Other | Git version-control, Latex, Linux OS, HPC slurm |